Practical 7: Spatial Data

Getting to grips with Geo-Data using Geopandas

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Last week we did some initial processing on the Inside Airbnb listings data, focussing on its *numeric* properties. This week we are going to focus on the *spatial* properties of the data set.

> Tip
>
> It makes life a lot easier if you gather all of the library import commands and configuration information (here having to do with `matplotlib`) in the first exectuable code block in a notebook or script. That way it's easy for you for others to see what what it is necessary to have installed before getting started!.

1 Preamble

```
from pathlib import Path
import numpy as np
import pandas as pd
import geopandas as gpd
import matplotlib.cm as cm
import matplotlib.pyplot as plt

import matplotlib
import matplotlib.font_manager
```

2 Reading Geo-Data

> ℹ Connections
>
> We're building on the work done in previous practicals with a particular nod to the lecture on Data to create some useful functions that we can call on at-need to improve the ease of doing data analysis.

I find GeoPackages and GeoParquet to be by far the easiest way to distribute geo-data now: they both use a single file in a database-like format and include the projection information by default. In some cases QGIS can even embed information about rendering style!

We're going to do something *similar* to the `get_url` in order to download the file to our hard drive and save it there. The improvement is that we'll check to see if the file already exists and, if it does, return that so that you can don't have to keep downloading the same data week after week.

You'll need to add the documentation yourself and I've left a few `??` to challenge you.

2.1 Caching Remote Data

We don't want to continually download data over the Internet: 1) because it's not nice to whoever is hosting the data; and 2) because it requires you be online in order to run your code. So if we can 'cache' the data locally so that it's only downloaded once this makes life much, much easier.

I've used the Numpy-style comments here, but the Google-style also look good in this context and all styles of answer are acceptable so long as they *work*. See overview of commenting styles on DataCamp.

> Tip
>
> Use this as an opportunity to improve your ability to read code and to learn through documentation.

> ⚠ Difficulty level: Moderate

```
from pathlib import Path
from requests import get
from functools import wraps

def check_cache(f):
    @wraps(f)
    def wrapper(src:str, dst:str, min_size=100) -> Path:
        if src.find('?') == -1:
            url = Path(src)
        else:
            url = Path(src[:src.find('?')])
        fn  = url.name  # Extract the filename
```

```
        dsn = Path(f"{dst}/{fn}") # Destination filename
        if dsn.is_file() and dsn.stat().st_size > min_size:
            print(f"+ {dsn} found locally!")
            return(dsn)
        else:
            print(f"+ {dsn} not found, downloading!")
            return(f(src, dsn))
    return wrapper

@check_cache
def cache_data(src:Path, dst:Path) -> str:
    """Downloads a remote file.

    The function sits between the 'read' step of a pandas or geopandas
    data frame and downloading the file from a remote location. The idea
    is that it will save it locally so that you don't need to remember to
    do so yourself. Subsequent re-reads of the file will return instantly
    rather than downloading the entire file for a second or n-th itme.

    Parameters
    ----------
    src : str
        The remote *source* for the file, any valid URL should work.
    dst : str
        The *destination* location to save the downloaded file.

    Returns
    -------
    str
        A string representing the local location of the file.
    """

    # Create any missing directories in dest(ination) path
    # -- os.path.join is the reverse of split (as you saw above)
    # but it doesn't work with lists... so I had to google how
    # to use the 'splat' operator! os.makedirs creates missing
    # directories in a path automatically.
    if not dst.parent.exists():
        dst.parent.mkdir(parents=True, exist_ok=True)

    # Download and write the file
    with dst.open(mode='wb') as file:
        response = get(src)
        file.write(response.content)

    print(' + Done downloading...')

    return dst.resolve()
```

Question

```
from pathlib import Path
from requests import get
from functools import wraps

def check_cache(f):
    @wraps(f)
    def wrapper(src:str, dst:str, min_size=100) -> Path:
        url = Path(src)
        fn  = url.name
        dsn = Path(f"{dst}/{fn}")
        if dsn.is_file() and dsn.stat().st_size > min_size:
            print(f"+ {dsn} found locally!")
            return(dsn)
        else:
            print(f"+ {dsn} not found, downloading!")
            return(f(src, dsn))
    return wrapper

@check_cache
def cache_data(src:Path, dst:Path) -> str:
    """Downloads a remote file.



    Returns
    -------
    str
        A string representing the local location of the file.
    """

    # Create...
    if not dst.parent.exists():
        dst.parent.mkdir(parents=True, exist_ok=True)

    # Download and write the file
    with dst.open(mode='wb') as file:
        response = get(src)
        file.write(response.content)

    print(' + Done downloading...')

    return dst.resolve()
```

```
print('```')
print('# help(cache_data)')
help(cache_data)
print('```')
```

```
# help(cache_data)
Help on function cache_data in module __main__:
```

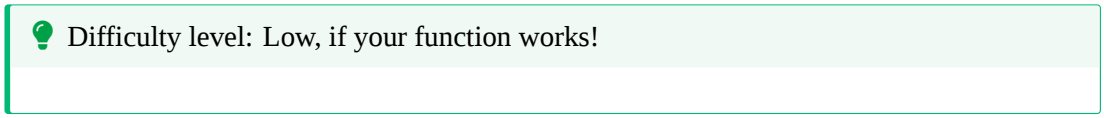
```
cache_data(src: pathlib.Path, dst: pathlib.Path) -> str
    Downloads a remote file.

    The function sits between the 'read' step of a pandas or geopandas
    data frame and downloading the file from a remote location. The idea
    is that it will save it locally so that you don't need to remember to
    do so yourself. Subsequent re-reads of the file will return instantly
    rather than downloading the entire file for a second or n-th itme.

    Parameters
    ----------
    src : str
        The remote *source* for the file, any valid URL should work.
    dst : str
        The *destination* location to save the downloaded file.

    Returns
    -------
    str
        A string representing the local location of the file.
```

2.2 Downloading the Data



2.2.1 Geopackages

Use the function above to download and cache the GeoPackage files found on GitHub for Boroughs, Water, and Greenspace, then pass the output of these to GeoPandas. If you have been having trouble downloading files from GitHub, then use the understanding of the function developed above to download the file manually and place it where this function expects to find it!

Question

```
ddir  = Path('data/geo') # destination directory
spath = 'https://github.com/jreades/fsds/blob/master/data/src/' # source path

boros = gpd.read_file( ??(spath + 'Boroughs.gpkg?raw=true', ddir) )
water = gpd.read_file( ??(spath + 'Water.gpkg?raw=true', ddir) )
green = gpd.read_file( ??(spath + 'Greenspace.gpkg?raw=true', ddir) )

print('Done.')
```

2.2.2 Parquet

Let's re-use our `cache_data` function to download and save the full Inside Airbnb data set. Again, if you have trouble with downloading via code, use your understanding of the function to work out where to save your own copy of this file so that the function works as expected.

> Difficulty level: Low

```
host = 'http://orca.casa.ucl.ac.uk'
ymd  = '20250615'
city = 'London'
file = f'{ymd}-{city}-listings.parquet'
url  = f'{host}/~jreades/data/{file}'
```

Question

```
# your code here
df = pd.read_parquet( ??(??, Path('data/raw').resolve()) )
print(f"Data frame is {df.shape[0]:,} x {df.shape[1]}")
```

You should see that the file was 'not found' so 'downloading' happened and then the size of the data frame was printed out.

3 'Writing' Geo-Data

Of course, you will also often encounter geo-data that is not yet 'geographically enabled'; the two most frequent contexts for this are:

1. The data represents points and is provided with latitude and longitude (or similar) as separate columns in a non-geographic data set.
2. The data represents polygons but is provided *separately* from the polygons themselves and so cannot be shown on a map without being 'joined' to the geography first.

We'll tackle each of these eventually, but for now we're going to focus on the first option.

3.1 Creating a GeoDataFrame

> Difficulty Level: Low

Right, we're finally there! We need to convert our coordinates into some kind of geo-data. GeoPandas offers two ways to do this: the original way using `zip` and a new utility method called `points_from_xy`. Here's the old way **which still works but shouldn't be used**:

```
from shapely.geometry import Point
gdf = gpd.GeoDataFrame(df,
```

```
                    geometry=[Point(x,y) for x, y in zip(df.longitude,df.latitude)])
```

The old way is interesting because it's very explicit about what you're doing (creating a list of Points that will be come a new GeoSeries), but it does not allow you to set a projection, unlike the new approach with the 'helper function' from the module **which is what you should use**:

```
gdf = gpd.GeoDataFrame(df,
      geometry=gpd.points_from_xy(df.longitude, df.latitude, crs='epsg:4326'))
gdf.to_parquet(Path(f'data/geo/{ymd}-{city}-listings.geoparquet').resolve())
```

```
print(type(gdf))
print(type(gdf.geometry))
gdf.geometry.iloc[1:5]
```

3.2 Saving Geo-Data Locally

3.2.1 Think Storage Formats

We want to save the InsideAirbnb GeoDataFrame to the 'geo' directory, but first let's see what file formats are supported.

Format	Advantages	Disadvantages
Shapefile	Wide support	Easy to break via loss of individual file element and not very efficient format
GeoJSON	Human-readable	Highly inefficient storage format
GeoPackage	Lightweight spatial database with benefits of indexing	Lightweight spatial database with limits on data types supported
GeoParquet	Highly compressed columnar database	Limited GIS support (except QGIS), and DuckDB has issues.^[You can install the Spatial Extension.

GeoPackages are really well-supported by QGIS: you can even embed multiple layers with different style information so that your final analysis is easy to distribute; however, they also carry quite a bit of overhead that makes them inefficient for distributing smaller data sets, while *also* not supporting the full spectrum of Pythonic data structures such as categorical data or lists.

GeoParquet is an extension of the Parquet format. For working with large data sets this has revolutionised my workflow: getting excited about a columnar database might seem a bit… nerdy… but it's been transformative for many data scientists. First, because the data set is columnar you only read in the data that you need, so reading Parquet file is blindingly fast. Second, you can stream data from a Parquet file over the Internet, so that means you even gain these advantages reading remote files. Third, you have full support for Python data types, including (up to a point) objects. And, fourth, you can treat multiple Parquet files with the same layout as a single data set or mutiple Parquet files with different layouts as tables in a single database!

3.2.2 Specifying a Driver

Since there are *many* formats in which to save geo-data, rather than have multiple `to_format_x` methods, GeoPandas has tried to have *one* for local files (`to_file`). Normally, Geopandas will do the 'right thing' based on the extension: so if it sees `.shp` it will use the ESRI Shapefile driver, if it sees `.gpkg` it will use the Geopackage dirver, and the same for `.geojson`, etc.

There are two ways you *may* run into trouble:

1. If you are reading/writing a **URL** (e.g. https://github.com/jreades/…/src/Boroughs.gpkg?raw=true) where there is content *after* the `.gpkg` exten sion. If in doubt, specify the driver.
2. If you are reading/writing a **Parquet** file then you need to use `to_parquet()` and `read_parquet()` for compatibility with Pandas.

So why didn't we need to do with the `cache_data` function earlier as well? Well, this was a side-benefit of using the standard URL library: it *automatically* stripped off the query string (`?raw=true`) when I asked it for the file name, so we saved the file locally as a GeoPackage with `.gpkg` extension, which means that GeoPandas could read it without any problems.

So the following two bits of code are equivalent:

```
# Should be fine for local files but...
boros.to_file('test.gpkg')
# This is safer if working across multiple computers/the Internet
boros.to_file('test.gpkg', driver='GPKG')
```

You *may* find that saving to a GeoPackage fails with a `TypeError` because of the presence of **Categorical data** in the geo-data frame. This behaviour seems to have changed recently so that it may no longer be an issue, but a 'geoparquet' file will not have this problem, so that's another reason to use it!

3.3 Spatial Indexing

It's also worth comparing the output of a point with the output of a polygon or multi-polygon because you may well come across data in formats (e.g. WKT) resembling both of these in real data sets and they *can* be read as well. Notice too that we can use `loc` and `iloc` accessor methods to pull individual points and polygons out of a GeoDataFrame!

A polygon:

```
# So this is a multi-polygon boundary
boros.geometry.iloc[1]

# Print out first 399 characters of the string
print(str(boros.geometry.iloc[1])[:399] + "...")
```

A point:

```
# Print out the object as a string
print(boros.geometry.centroid.iloc[1])
```

```
# And here's a point
boros.geometry.centroid.iloc[1]
```

So each element of this Series has text indicating the type of shape the geometry applies to (e.g. *POLYGON*) followed by a bunch of numbers. These numbers are truncated here just to make things a little more legible.

4 Checking the Data

4.0.1 Checking Numeric Data

Before we mindlessly convert the parquet data to mappable geo-data it might make sense to sanity-check it. GeoPandas has a `total_bounds` method that gives us the bounding box for a GeoSeries, but how would we do that in Pandas?

> **Tip**
>
> Think about what the 'total bounds' (or 'envelope') of a point data set is. You have already seen the pandas functions you'll need to *find* these…

> **Difficulty level: Moderate**

Question

```
print(f"The bottom-left corner is {??}, {??}")
print(f"The top-right corner is {??}, {??}")
```

Your answer should produce the following:

- The bottom-left corner is -0.5, 51.3
- The top-right corner is 0.3, 51.7

4.0.2 Checking the Projection

> **Difficulty level: Low**

Print out the projection of each GeoDataFrame (the water, borough, greenspace, and InsideAirbnb data) using a for loop.

Question

```
for g in [??]:
    print(g.??)
```

You should see that three are in the EPSG:27700 CRS which is a common one for analysis using GB data, but the InsideAirbnb data is not.

4.0.3 Reprojecting the Data

> Difficulty Level: Low

Let's view the CRS information for the InsideAirbnb data in its original projection…

```
print(gdf.geometry.crs)
print(gdf.total_bounds)
```

… and reproject this into the `OSGB1936/BNG` CRS:

Question

```
gdf = gdf.??(??) # There is no 'in_place=True' option here.
print(gdf.geometry.crs)
print(gdf.total_bounds)
```

The bounding box is: 504112.02547, 156900.30057, 559264.71912, 199928.86103

5 Simple Mapping

Behind the scenes, GeoPandas is using `matplotlib` to render the map, so let's play with the colours to get the *start* of something map-like. You will want to look both at how to make maps in GeoPandas and at the different ways to specify colours in Matplotlib. For the greenspace map you are looking for information about tuples… which can have three or four elements.

5.1 Making Plots

We'll see how you control figure-making more effectively later, but for now let's just see what they look like using GeoPanda's plotting functionality. You should output three plots from the loop.

Question

Can you plot the 'boros', water, and green space in a loop?

```
for ??:
    ??.plot()
```

5.2 Work Out the Colour Scheme

Difficulty level: Moderate

Caution

R and Python take *very* different approaches to plotting. Do *not* think of Python's output as being 'maps' in the GIS sense, they are composed of 'patches' of color on abstract 'axes' that can use any arbitrary coordinate space. So colours are 'really' triplet (or quadruplet if you have alpha-blending transparency) values in the range 0.0-1.0. Annotations are then added in similarly abstract fashion.

I'd **suggest** the following colour scheme as a way to test out different ways of specifying colour (though anything you like is fine so long as you manipulate the colours):

- The boroughs can have red edges and white fill with a thick edge.
- The water should have no edges and XKCD Lightblue fill.
- The greenspace should have edges and faces specified using different 'alpha blending' (i.e. transparency) levels.

5.2.1 Boroughs

By way of a hint, matplotlib uses `edgecolor` and `facecolor` for controlling 'patches' (which is what polygons are considered), but the thicker-than-default line-width is specified differently (you'll need to look this up). So the intention is:

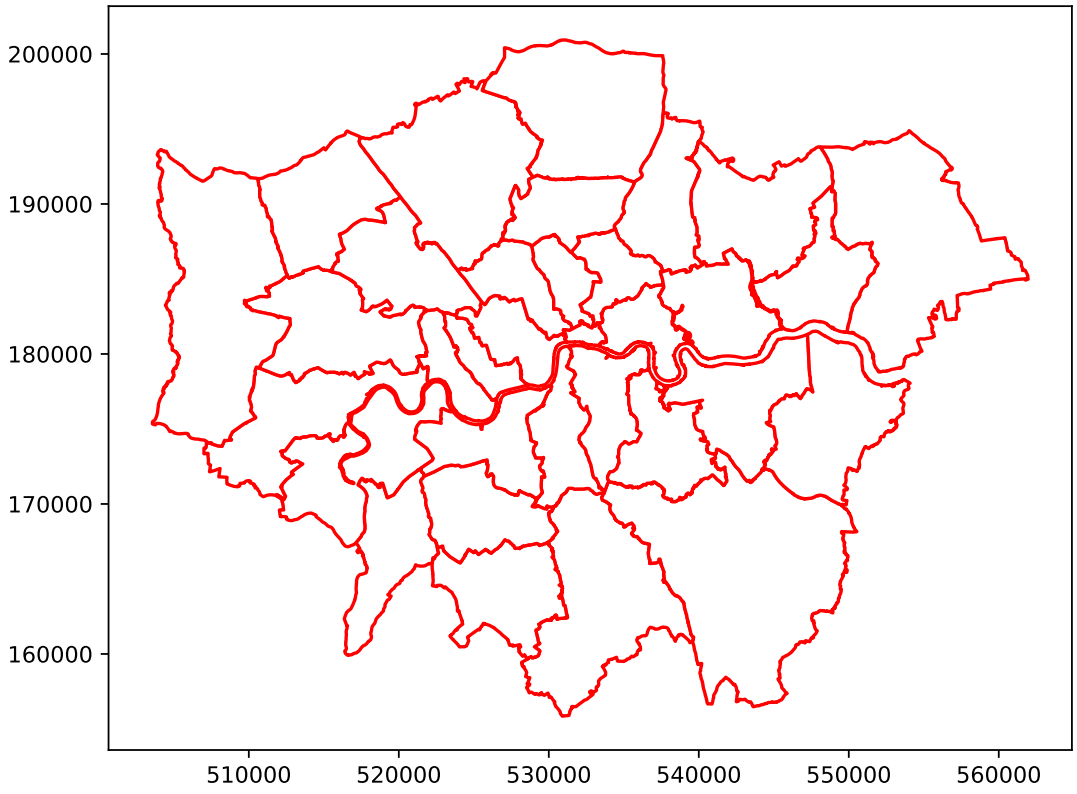
1. Thick red borough borders, and
2. White fill colour.

Just to drive home how different this is from R, you can find the answer to question 1 on the page for bar plots.

Question

```
boros.plot(??, figsize=(8,6))
```

Your plot should look similar to this:



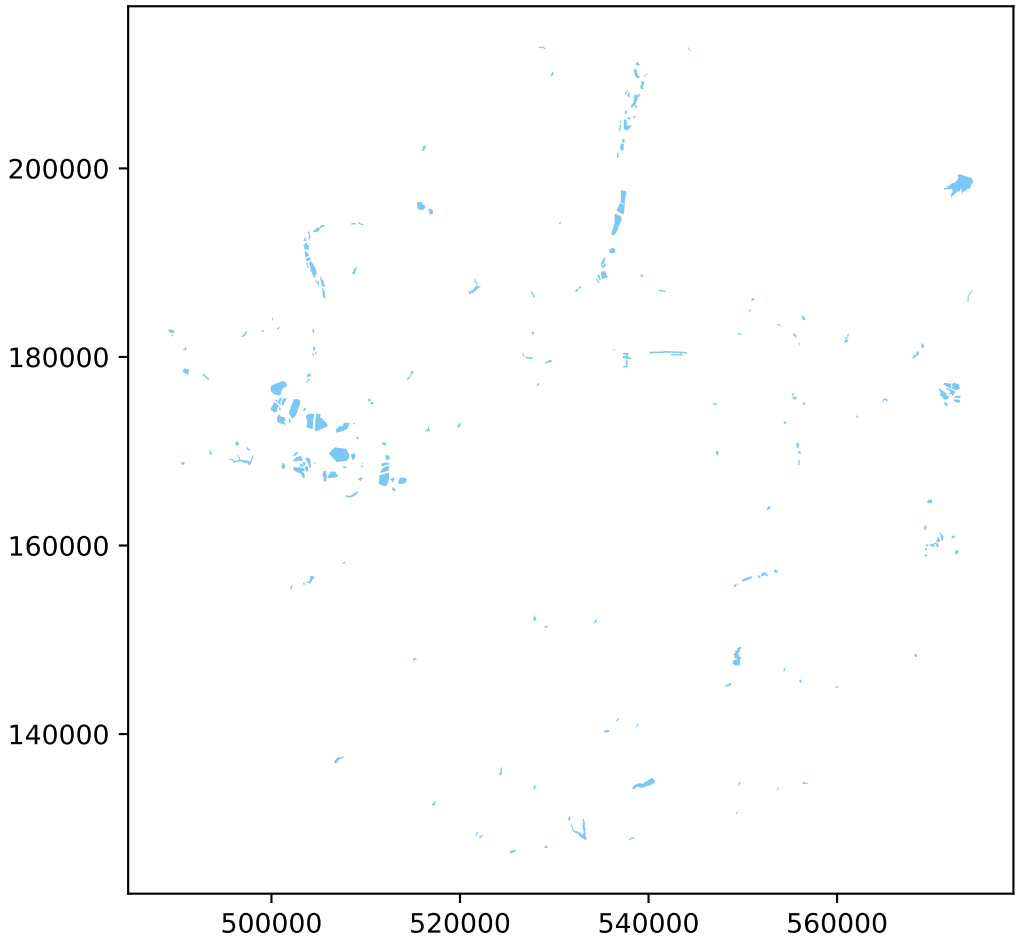
5.2.2 Water

The process is the same as above, but I'd like you to work out how to specify: 1. *No* color for an edge, and 2. An XKCD color for the face.

Question

```
water.plot(??, figsize=(8,6))
```

Your plot should look similar to this:



5.2.3 Greenspace

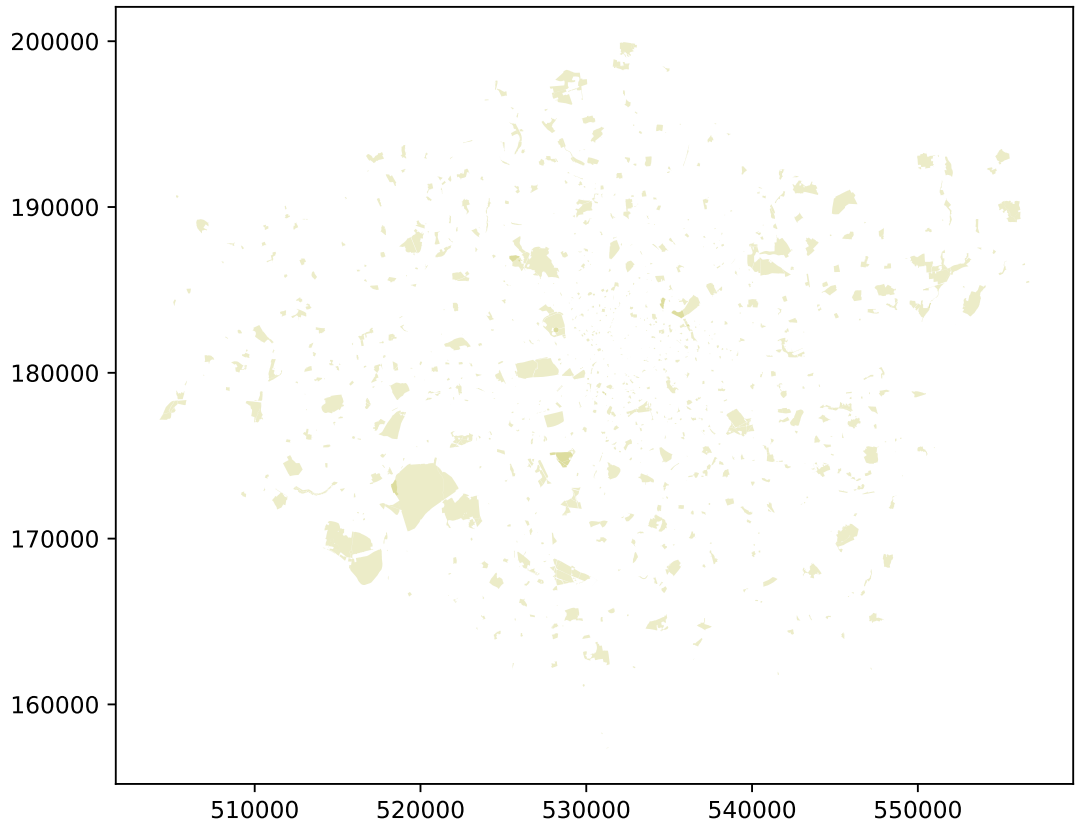
The process is *also* the same as above, but I'd like you to work out how to specify colours and transparency using RGBA (red-green-blue-alpha transparency) tuples. So we're looking for:

1. No edge color.
2. A partially transparent green specified as a 'tuple' (4 numbers in parentheses in the range 0.0-1.0).

Question

```
green.plot(??, facecolor=(??), figsize=(8,6))
```

Your plot should look similar to this:



5.3 Combining Layers

> **Warning**
>
> R and Python take *very* different approaches to plotting. Do *not* think of Python's output as being 'maps' in the GIS sense, they are composed of 'patches' of color on abstract 'axes' that can use any arbitrary coordinate space. So colours are 'really' numerical triplets (or quadruplets if you have transparency as well) in the range 0.0-1.0. Annotations are then added in similarly abstract fashion.

Now that we've got our layers looking roughly how we want them, it's time to combine them. This is also reliant on `matplotlib` and basically involves plotting items to *shared axes* which is done by passing in `ax=<axis object>` to each `plot(...)`. By convention, if you only have a single figure (e.g. a single map) then you create an axis object and name it `ax` so you will see a lot of `ax=ax` code in graphing libraries, but `=ax` is just saying 'assign to the axis object that I created'.

Since the axes are how you control what is shown, see if you can find out by Googling how to set the x- and y-limits on the map so that it shows only London and trims out the much larger area of water that is outside of the Greater London Authority. **As a rough guideline, this has the Easting range 501,000 to 563,000, and the Northing range 155,000 to 202,000.**

You can set these limits before or after you start adding layers to the 'map', but it's probably easier conceptually to add them after with the idea of 'zooming in' on the features of interest. It's also easier to debug since you can start by seeing if you can plot the elements at all, and *then* add the limits to zoom.

So the steps are:

1. Write the code to plot every image on the same set of axes (I've given you something to get started).
2. Google how to set the limits of the map and then use the ranges I've offered above.
3. Work out how to change the width of the edges for the boroughs layer.
4. Save it somewhere local so that you could, say, load it into a Markdown file!

> Tip
>
> This is a first pass at a map, over the next few weeks we'll see how to add things like axis labels and titles to make it more 'map-like'. We don't have quite the built-in functionality of `ggplot` alas, but Python is advancing very quickly in this area. There is even an implementation of ggplot in Python, but it's functionality is more limited. In fact, there's more than one…

> Difficulty level: Hard

Question

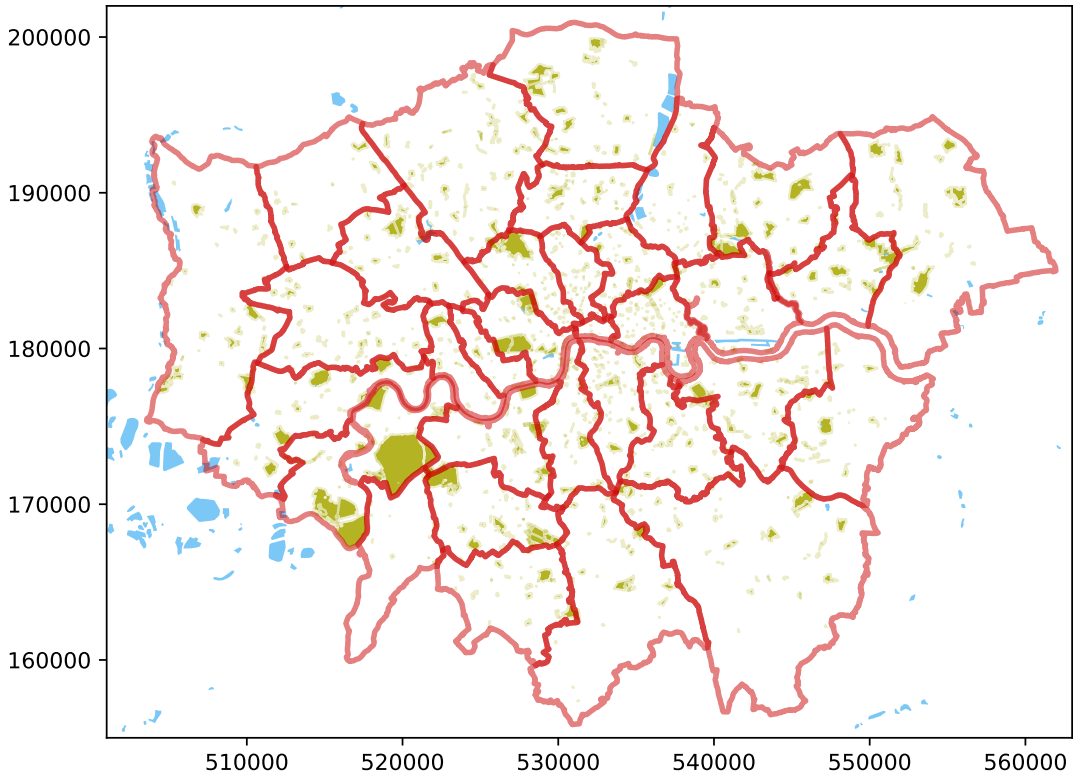
```
# Creates a new figure with specified number of
# subplots (we'll see more of this later) and
# and the specified size (in inches by default).
fig, ax = plt.subplots(1,1, figsize=(8,6))

# Plot all three GeoPackages to the same axes
water.plot(??, ax=ax)
green.??
boros.??

# Set the x and y limits


# Save the image (dpi is 'dots per inch')
Path('img').mkdir(parents=True, exist_ok=True)
plt.savefig(Path('img/My_First_Map.png'), dpi=150)
```

You may wish for a different look, but here's one version of the output:



5.4 Choropleth Plots

Difficulty Level: Low

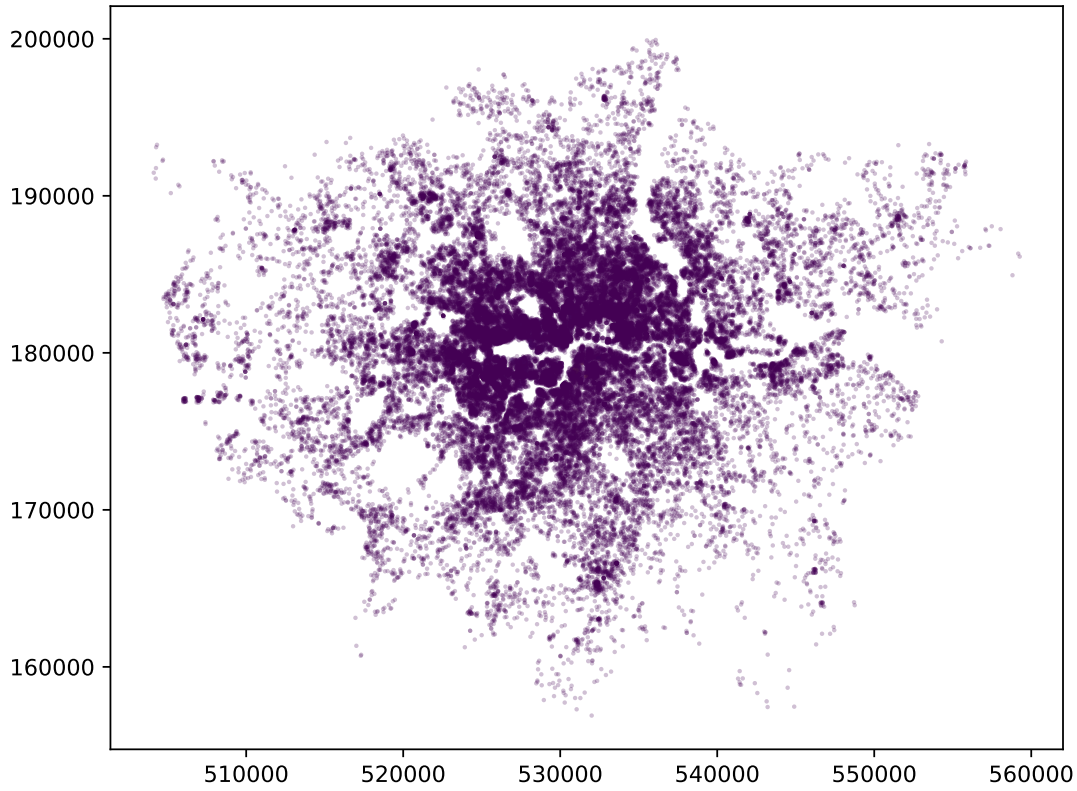
Now that we've converted the InsideAirbnb data to a GeoDataFrame, we can plot it, reproject it, etc.

See if you can work out how to plot the points coloured by their price using the appropriate BNG projection.

Question

```
gdf.to_crs(??).plot(column=??, cmap=??, alpha=0.25, markersize=1, figsize=(8,6));
```

Using the Viridis colourmap I get the following:



5.4.1 Work out the Data Range

Difficulty level: Low

As we saw above with the point-plot, in its original form the pricing data will not reveal much of interest because of the range of the data. However, using *transformations* we can manipulate the data to increase its tractability for analysis. We saw several transformations last week (*z*-score, natural log, etc.).

Let's start by getting a feel for the full data set in terms of the range of prices that it contains:

Question

```
print(f"The range of price is ${??:,.2f} to ${??:,.2f}")
print(f"The mean and median of the price are ${??:,.2f} and ${??:,.2f}")
```

Notice the neat little comma-separated thousands in there? That's fairly easy to do in English, but to use a thousands separator common to another language you would need to do something a little more tricky.

5.4.2 Inheritance!

We already know that GeoPandas *inherits* functionality from Pandas, but let's formalise this…

First, let's check what class of object `gdf` is using the `isinstance` function:

```
# Is gdf a GeoDataFrame object?
if isinstance(gdf, gpd.GeoDataFrame):
    print("\tI'm a geopandas data frame!")

# Is gdf *also* a DataFrame object?
if isinstance(gdf, pd.DataFrame):
    print("\tI'm a pandas data frame!")
```

5.4.3 Benefiting from Inheritance

That result means that we can also investigate the data using, for instance, a pandas histogram and here are two ways to look at the pricing data:

```
# Oooooh, let's use a *pandas* method here
gdf.price.plot.hist(bins=200, xlim=(0,1000), figsize=(8,3));
```

```
# Oooooh, let's use a *pandas* method here
gdf[gdf.price < 1000].price.plot.hist(bins=200, xlim=(0,1000), figsize=(8,3));
```

Question

Can you explain why these two plots look very different?

Notice how we've used our GeoDataFrame *as if* it's a plain old DataFrame here? That's the miracle of Object-Oriented Design: we can do *anything* we would with a regular Pandas `df` as we do with a GeoPandas `gdf` because GeoPandas *inherits* all the methods of its parent super-class.

We can see that there's very little data above (at a guess) about $1,000, but at this scale it's hard to tell. We've already seen that you can use axes limits to adjust the display of a map, but the *same* technique applies to plain old plots because they're fundamentally the *same thing* in that they're both `matplotlib` outputs.

You can do the same thing with a `box`plot:

Question

```
ax = gdf.price.plot.??(vert=False, figsize=(8,3))
ax.??
```

More complex formatting is also possible if you really know your pandas and your matplotlib:

```
gdf.price.plot(kind='box', vert=False,
               color=dict(boxes='r', whiskers='r', medians='r', caps='r'),
               boxprops=dict(linestyle='-', linewidth=1.5),
               flierprops=dict(linestyle='-', linewidth=1.5),
               medianprops=dict(linestyle='-', linewidth=1.5),
               whiskerprops=dict(linestyle='-', linewidth=1.5),
               capprops=dict(linestyle='-', linewidth=1.5),
               showfliers=False, grid=False, rot=0);
```

5.5 Truncate and Transform

> Difficulty level: Hard

5.5.1 Working it Out

Anyway, drawing on everything we've seen over the past couple of weeks I'd like you to:

1. Try to take the natural-log of the price (*hint*: use `numpy`) and assign to a new Series called `lnprice`.
2. Work out what the error means.
3. Work out how to fix the error and *then* repeat step 1.
4. Work out how many rows were affected.
5. Report on the new min/max values.
6. Work out if other outliers need to be removed (use code from above).
7. Remove outliers and then continue with your work…

```
# Use this as a 'scratch space' to work out what's needed below...
```

Question

```
print(f"gdf has {gdf.shape[0]:,.0f} rows.")

# ---------- Do the processing -------------
# This will throw an error 'as is'
gdf['lnprice'] = np.log(gdf.price)

# ---------- Check effects -----------
print(f"The range of ln(price) is {gdf.lnprice.min():,.4f} to {gdf.lnprice.max():,.4

gdf.lnprice.plot(kind='box', vert=False,
             color=dict(boxes='r', whiskers='r', medians='r', caps='r'),
             boxprops=dict(linestyle='-', linewidth=1.5),
             flierprops=dict(linestyle='-', linewidth=1.5),
             medianprops=dict(linestyle='-', linewidth=1.5),
             whiskerprops=dict(linestyle='-', linewidth=1.5),
             capprops=dict(linestyle='-', linewidth=1.5),
             showfliers=False, grid=False, rot=0);
plt.title("Ln(Price) (Outliers not shown)")
plt.show()
```

5.6 Plot Options

Now plot the ln(price) as a chloropleth using:

1. A figure size of 9 x 6
2. A marker size of 0.25

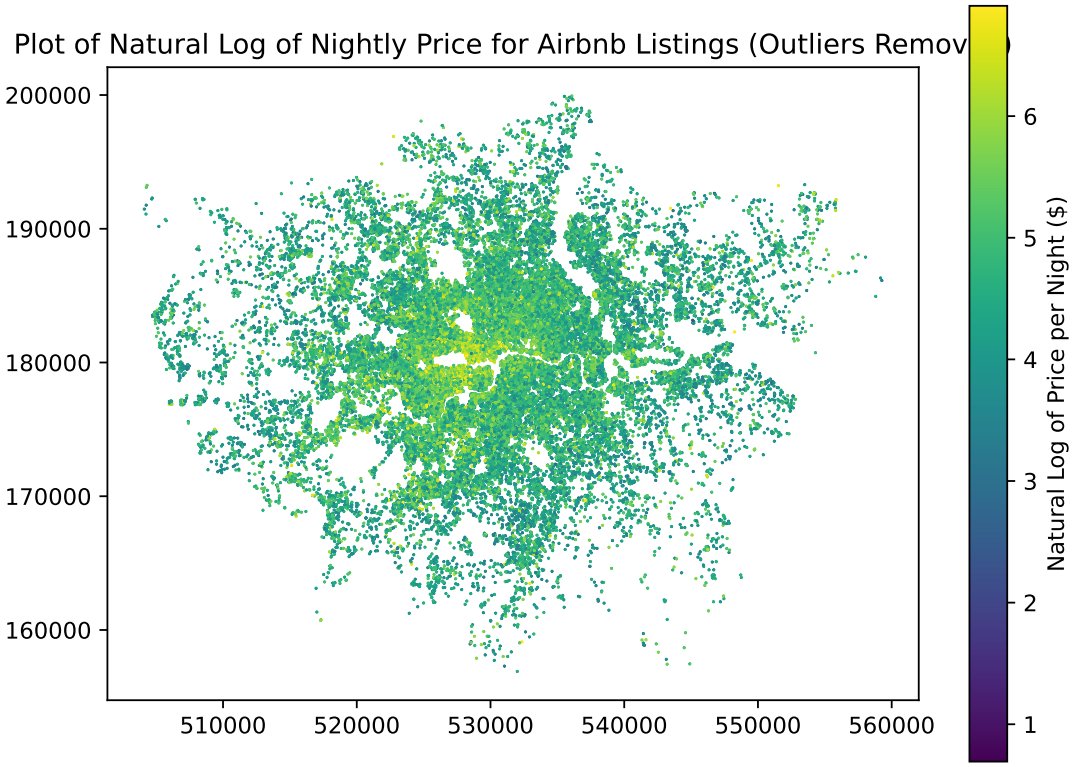
3. The Viridis colourmap
4. A legend
5. A legend label of ‘Natural Log of Price per Night ($)’

I’d suggest referring to the documentation.

Question

```
ax = gdf[ ?? ].plot(figsize=??, marker='*', markersize=0.25,
         column=??, cmap=??,
         legend=??, legend_kwds=??);
ax.set_title("Plot of Natural Log of Nightly Price for Airbnb Listings (Outliers Rem
```

You should get something like:



5.7 Zooming In/Out

> Difficulty Level: Low

That’s a little hard to see, let’s try zooming in on Central London! Very roughly, let’s call that an Easting range of 525,000 to 535,000 and a Northing range of 178,000 to 185,000.

> Plotting
>
> We show one way to do this below (`f,ax = plt.subplots(...)`) because it gives you greater control, but `gdf.plot()` *can* return an axis object (`ax = gdf.plot(...)`) that

gives you the same kind of access… but with a bit more ‘faff’.

Question

```
# Note this new f,ax syntax and that we then
# pass ax=ax to gdf.plot -- this has to do with
# where and how things are plotted.
gdf = gdf[ (gdf.lnprice > 0) & (gdf.lnprice < np.log(1000)) ]
f,ax = plt.subplots(1,1,figsize=(8,6))
gdf.plot(ax=ax, marker='*', markersize=0.25,
         column='lnprice', cmap='viridis',
         legend=True, legend_kwds={'label':'Natural Log of Price per Night ($)'});
ax.set_title("Ln(Price/Night) for Airbnb Listings (Central London Detail)")
ax.??
ax.??
f
```

Your result should look something like this:



That’s a little better, but ideally we’d do more thinking about outliers…

5.8 Changing the Classification Scheme

⚠ Difficulty Level: Moderate (mainly computation time)

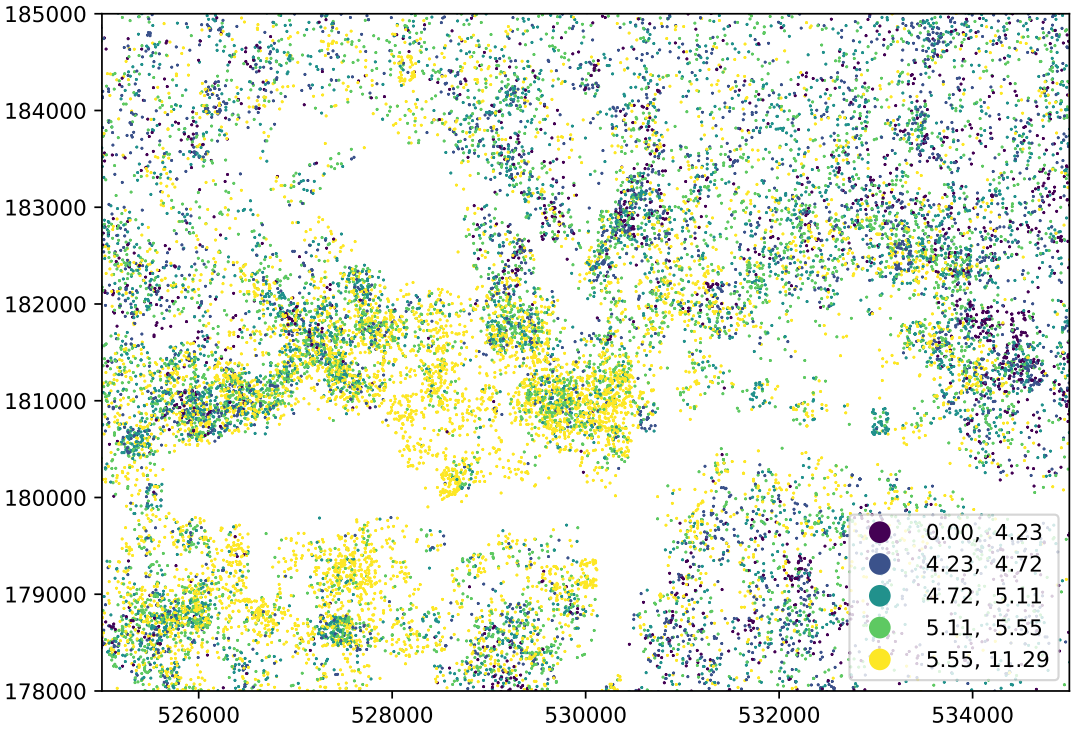
Let’s give this one last try using the quantiles classification scheme for Central London!

Question

```
f,ax = plt.subplots(1,1,figsize=(8,6))
ax = gdf.plot(marker='*', markersize=0.25,
         column='lnprice', cmap='viridis', ??, k=5,
         legend=True, ax=ax);
         # Note that the legend *label* had to go --
         # there are other ways to add it.

ax.set_xlim([525000,535000])
ax.set_ylim([178000,185000])
```

Your answer should look like:



6 Simple Web Maps

Difficulty Level: Hard

This is more for the sake of demonstrating Python's features than because it's part of my workflow, but what the heck, let's do it! We will create and embed a zoomable web map in the notebook; to do *that* we need to:

1. Calculate the bounds of the map using the min/max x and y coordinates above.
2. Calculate the centroid of the map from the bounds.
3. Set an appropriate zoom level.

If your work is going well, perhaps you may also want to experiment with different basemaps.

> **Tip**
>
> You can't use `round` here because it it could round up or down depending on what's closest and, consequently, cut off data on your map. So you'll have to look for two *other* functions that do this predictably (e.g. always rounding down, even if the value is 4.999999). However, those functions don't handle decimals like `round` does, so you need to think about how you could turn a number like 4.99 into a number that those functions *can* work with and then turn it *back* into the decimal…

Question

```
from math import floor, ceil

# Calculate min and max to *two* decimal places
xmin = ??
xmax = ??
ymin = ??
ymax = ??

# Print them to *3* decimal places to check they end in 0
print(f"{xmin:.3f}, {xmax:.3f}, {ymin:.3f}, {ymax:.3f}")

# Calculate the centre of the map
yctr = ??
xctr = ??

# Print this two ways to see an intriguing issue
print(f"{xctr:.5f}, {yctr:.5f}")
print(xctr, yctr)
```

You should end up with something like:

```
-0.500, 0.300, 51.290, 51.690
-0.10000, 51.49000
-0.09999999999999998 51.489999999999995
```

You'll see *why* this happens in the answer.

If you've managed the calculations above, then this code should simply run!

```
from ipyleaflet import Map, basemaps, basemap_to_tiles, Rectangle, projections

# Note the basemap can be easily changed
watercolor = basemap_to_tiles(basemaps.OpenStreetMap.HOT)

m = Map(layers=(watercolor, ), center=(yctr, xctr), zoom=8)

rectangle = Rectangle(bounds=( (ymin, xmin), (ymax, xmax) ),
    crs=projections.EPSG4326
)
```

```
m.add_layer(rectangle)

display(m)
```

With luck, that will give you a dynamic, scrollable map!

7 Bringing it All Together

```
# We change fonts the hard way in this notebook...
# but you can also do this to change the default
# font everywhere in one go:
# matplotlib.rcParams['font.family'] = "Liberation Sans Narrow"
fontname = "Roboto Slab"

matplotlib.font_manager.FontManager().findfont(fontname, fontext='ttf', rebuild_if_m
```

```
Matplotlib is building the font cache; this may take a moment.
```

```
'/Users/jreades/Library/Fonts/RobotoSlab-VariableFont_wght.ttf'
```

Difficulty Level:

To give a bit of a show of how we can put it all together try to get this working for something approximating a half-decent map.

Note that the following command seems to produce a lot of warnings for me: reading closely, I think it's about changes to Python with 3.12 and higher which have triggered warnings to do with syntax that *was* perfectly legitimate with Python 3.10 but is now more strongly deprecated.

```
import pysal as p
```

Question

```
import pysal as p
import mapclassify as mc
import palettable.matplotlib as palmpl
from legendgram import legendgram

xmin = 525000
xmax = 540000
ymin = 176000
ymax = 186000

# Reload the data because we've filtered it by price, not geography
gdf = gpd.read_parquet(Path(f'data/geo/{ymd}-{city}-listings.geoparquet').resolve())
```

```
# We create a temporary data frame here because we want
# the 'bins' to be created using only the data on the
# map. Otherwise, we'd have a distribution on the map
# that differed from the one in the legendgram and the
# one used to calculate the breaks in the first place!
tgdf = gdf[
    (gdf.geometry.x > xmin) &
    (gdf.geometry.x < xmax) &
    (gdf.geometry.y > ymin) &
    (gdf.geometry.y < ymax) &
    (~gdf.price.isna()) &
    (gdf.price > 0) &
    (gdf.price < 1000)].copy()

# Here we use Mapclassify to calculate quantiles
# (k=4) using the original price. You could use
# any Mapclassify scheme at this point, though
# note that for Fisher Jenks you might want to use
# the 'Sampled' version to speed things up a bit.
q = mc.??(tgdf.price.values, ??)

# We then write these binned values *back* on to the data
# frame so that we can use them with the GDF plot function.
tgdf['bins'] = q.??

# Set up the figure with its 'basemap'
f,ax = plt.subplots(figsize=(8,6))
green.plot(edgecolor=(0.7, 0.7, 0.14, 0.25), facecolor=(0.7, 0.7, 0.14, 0.25), zorde
water.plot(edgecolor="none", facecolor='xkcd:lightblue', zorder=2, ax=ax)
boros.plot(edgecolor=(0.8, 0, 0, 0.5), facecolor='none', linewidth=2.5, zorder=3, ax

# Restrict the x and y axis to the data
# Notice it's hard-coded here *and* above
# this could be done *better* using a
# custom bounding box so that we could
# update both the tgdf and the display
# area at the same time.
ax.set_xlim([xmin,xmax])
ax.set_ylim([ymin,ymax])

ax.axis(??) # Don't plot the axes

# Plot the bins using a categorical legend instead
# of the price using a continuous legend.
tgdf.plot(column='bins', categorical=True,
         cmap='viridis', legend=True, marker='.', markersize=1.5, zorder=4, ax=ax)

# Set the title using a specified font, weight, and size
ax.set_title('London Airbnb Listings Price Per Night',
             fontdict={'fontsize':'20', 'fontweight':'3', 'family':fontname})  #prov
```

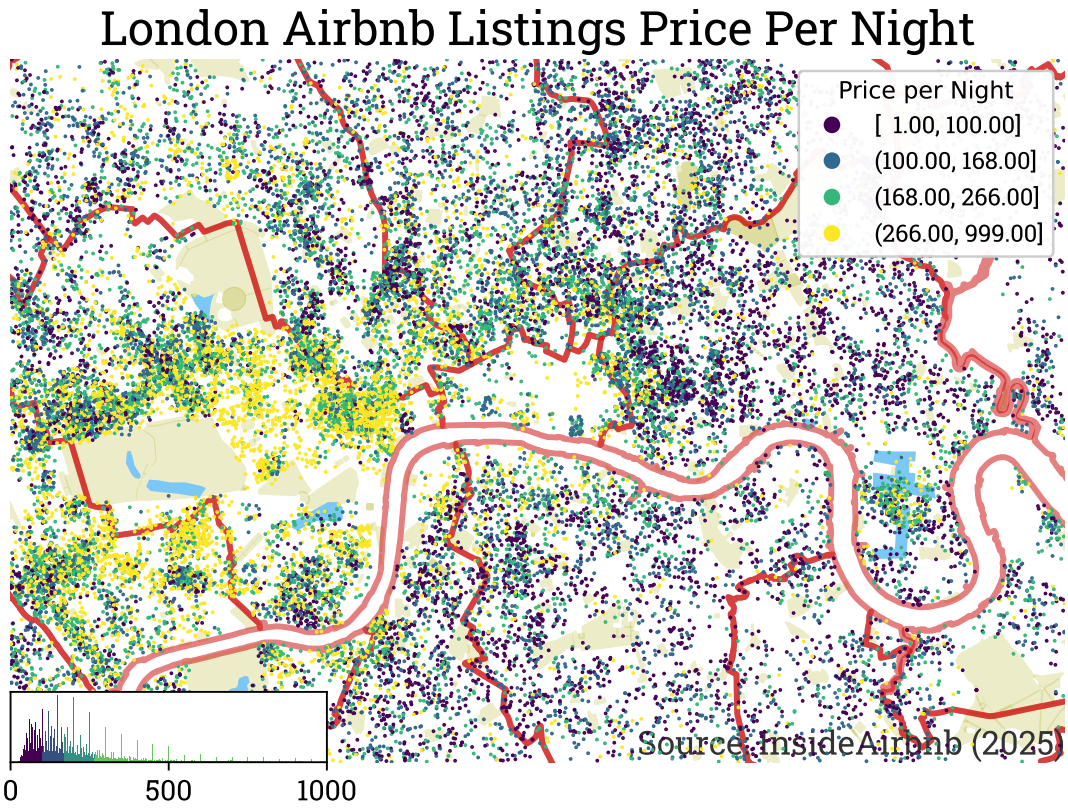
```
# This is where geopandas gets in the way -- the
# categorical legend doesn't work for us so we need
# to actually create the legend 'by hand' using this
# code... which first has to *find* the layer containing
# the data! Each layer is a 'patch collection', so we
# loop through the collections looking for the one whose
# z-order is 4 (which we set above to the data layer).
#
# I relied on this: https://stackoverflow.com/a/71419387/4041902
# to work out how to do this!
for c in ax.collections:
    # Find the layer with the data
    if c.get_zorder()==4:
        # *Now* we can create a legend... but we need to
        # first retrieve the colours from the layer. These
        # are returned as 'handles' and then we need to
        # associate these with the labels taken from the
        # Mapclassify object... Once we set that up, along
        # with fonts and such, we can add it as an 'artist'
        # to the figure.
        handles, _ = c.legend_elements(prop="colors")
        legend1 = ax.legend(handles, q.get_legend_classes(fmt='{:.2f}'),
                            loc="upper right", title="Price per Night",
                            prop={'size':'10', 'weight':'1', 'family':fontname})
        ax.add_artist(legend1)

# And don't forget to add a source!
a = ax.text(tgdf.geometry.x.max(), tgdf.geometry.y.min(), 'Source: InsideAirbnb (202
            horizontalalignment='right', verticalalignment='bottom',
            fontsize=14, fontweight=4, color='#333333', family=fontname)

# And this is a nice feature: show the distribution!
ax2 = legendgram(f, ax,
           tgdf.??, q.??, bins=round(gdf.price.max()/25),
           pal=palmpl.Viridis_4,
           legend_size=(0.3, 0.1),
           loc='lower left',
           clip=(0,1000),
           frameon=True
    )
# But we have to fix the font manually here
# for the legendgram too
for tk in ax2.get_xticklabels():
    tk.set_fontname(fontname)

#plt.savefig('Airbnb-price-all.png', dpi=150)
```

You should have something like this:



I'll be honest, I do find ggplot easier for making good-quality; this *is* more customisable overall, but it's also much more 'magical' in the sense of 'search for `matplotlib` examples that do what you want then copy+paste them and tweak' being the main way that most people get things working how they want.

Scalebars are awkward, and there's now a library that can help with this on GitHub that I've installed. But I'll leave that one to you.

> **Tip**
>
> You can find a lot of *possible* solutions in this Stackoverflow thread that *should* work without needing to install new libraries but I've not had a chance to test them each individually. You would undoubtedly want to put this in an external package and import it when needed rather than paste this code into every file. But you might find it easier to *test* the solutions by pasting. If you're looking for glory (and my gratitude) then working out which of these is most generalisable (i.e. would work with both lat/long *and* OSGB coordinates) would be quite the challenge!

7.0.1 Getting More Help/Applications

A great resource for more help and more examples is Dani Arribas-Bel's *Geographic Data Science* module: he has put all of his module practicals online (as we have too), and you might find that something that he does makes more sense to you than what we've done… check it out!

7.1 Credits!

Contributors:

The following individuals have contributed to these teaching materials: Jon Reades (j.reades@ucl.ac.uk), James Millington (james.millington@kcl.ac.uk)

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Potential Dependencies:

This notebook may depend on the following libraries: geopandas, pandas, matplotlib, seaborn